*Geophys. J. R. astr. Soc.* (1975) **43**, 1013–1016.

# Research Note

# Palaeomagnetism of the Breidden Hill (Palaeozoic) Inlier, Welsh Borderlands

**J. D. A. Piper and J. E. F. Stearn**

(Received 1975 May 7)

*Summary*

A palaeomagnetic survey of this inlier yields pole positions of Upper Ordovician age consistent with Ordovician data from elsewhere in Britain, and evidence for a magnetic field reversal.

## 1. Introduction

The Breidden Hill inlier straddles the Shropshire–Montgomeryshire border 15 km west of Shrewsbury and comprises a sequence of Ordovician shales and mudstones containing the graptolites Climacograptus *peltifer* and Nemagraphs *gracilis* (Watts 1925; Pocock & Whitehead 1948) belonging to the Caradoc (Lower Bala) division of the Upper Ordovician. These are overlain conformably by tuffs (palaeomagnetic sites 8 and 10, Fig. 1) and an andesite unit (Moel-y-Golfa) sampled at palaeomagnetic sites 9, 11 and 12. This andesite has been described as an intrusive body (Watts 1925) but includes autobrecciated horizons and is almost certainly a thick lava flow. The Ordovician rocks were intruded by dolerites including the Breidden Hill intrusion (palaeomagnetic sites 1–7 Fig. 1(a)) and a number of sills (site 13) prior to folding. They are overlain unconformably by Lower Silurian rocks which give a minimum age for the dolerite intrusions.

## 2. Palaeomagnetic results

Sites 1–7 in the Breidden Hill dolerite form an approximately equally-spaced sequence through 300 m of the intrusion. Magnetic properties of these sites show systematic variation through the intrusion with sites 3 and 4 having intensities of magnetization and susceptibilities between one and two orders of magnitude higher than sites near the margins of the intrusion (Fig. 1(b)). Site 4 also has the highest Köenigsberger $Q$ ratio of the collection correlating with highest stability to af demagnetization. The magnetic properties correlate with petrological properties: sites 1, 6 and 7 at the margins of the intrusion are highly altered with pyroxene thoroughly uralitized and feldspar sericitized; Sites 2 and 5 are less altered with some relict pyroxene and sites 3 and 4 are practically fresh. Content of opaque minerals tends to increase downwards through the intrusion from less than 5 per cent to about 10 per cent while the grains become progressively altered and smaller in size towards the marginal sites. Only sites 3 and 4 gave measurable Curie points with values of $550 \pm 15$ °C and little change in $J_s$ on cooling. That this represents a remanence held



Downloaded from https://academic.oup.com/gji/article/43/3/1013/819255 by Curtin University Library user on 26 May 2026

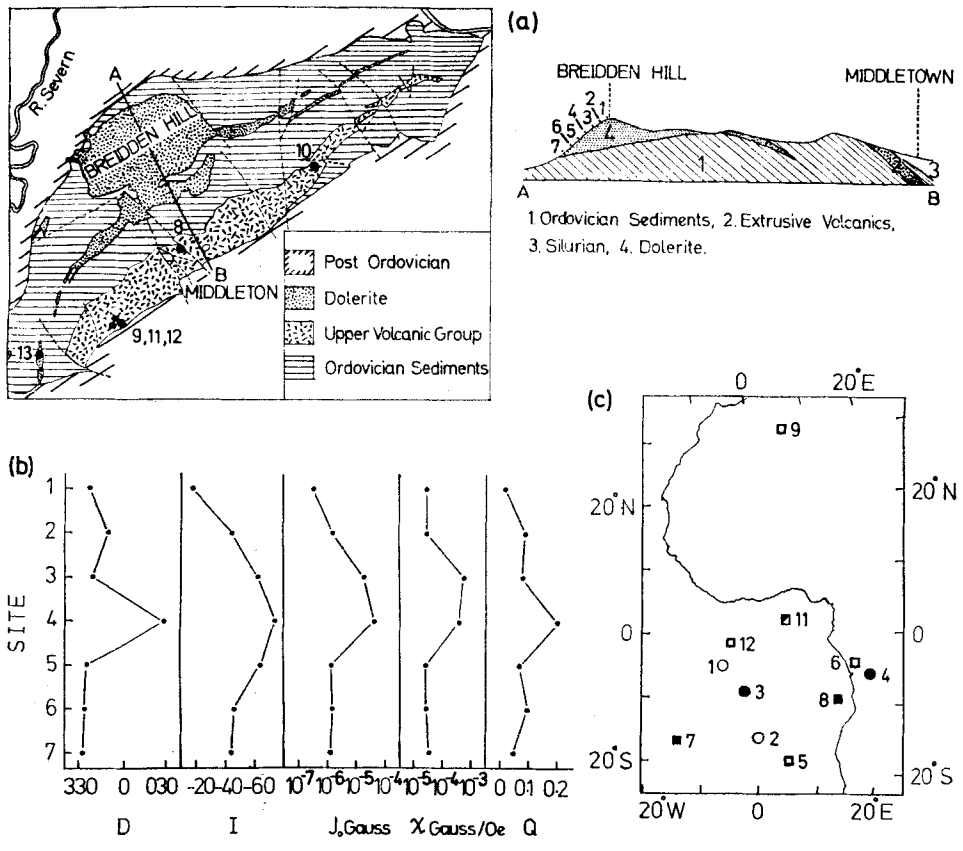



FIG. 1. (a) Distribution of sampling sites in the Breidden Hill inlier and through the Breidden Hill dolerite intrusion; geology is simplified after Watts (1925). (b) Distribution of magnetic properties through the Breidden Hill dolerite intrusion; since the precise altitude of each site within the intrusion is not known, they are equally spaced in this diagram. (c) Plot of pole positions derived from Ordovician rocks of England and Wales (this paper, Briden *et al.* 1973 and W. A. Morris, private communication). Numbered poles are from the Trefgarn Andesitic Series (1), Builth Volcanic Series (2), Eycott Group (3), Borrowdale Volcanic Series (4), Carrock Fell Complex (5), Breidden Hill intrusion (6), Caradocian tuffs, Breidden Hill inlier, site 8(7) and site 10(8), Caradocian andesite, Breidden Hill inlier, site 11(9), Minor intrusions of Ashgillian age Builth (11) Pre-Caradoc diorite dyke, Stile End (12). Closed symbols are poles and open symbols are antipoles.

principally by titanomagnetite is suggested by demagnetization characteristics, all sites showing sharp to moderate falls in moment with the first few hundred oersteds of treatment; at higher fields demagnetization curves typically become erratic.

All sites in the intrusion have the same polarity and combine after correction for regional tilt of the inlier (36° in a direction 147° E) to give a mean of $D = 341°$, $I = -51°$ and a palaeomagnetic pole at 193° E, 4° N. Variation in direction of magnetization through the intrusion is largely in inclination with progressive increase in this parameter towards the centre (Fig. 1(b)). Since this change parallels a progressive decrease in degree of low temperature hydrothermal alteration from the margins to the centre of the intrusion, it probably results from progressive remagnetization (possibly soon after consolidation and certainly in Ordovician times) with replacement of a primary thermal remanence, now only preserved in the interior of the intrusion, by a chemical remanent magnetization (CRM). The single site (13) from a dolerite sill at

Downloaded from https://academic.oup.com/gji/article/43/3/1013/819255 by Curtin University Library user on 26 May 2026

**Table 1**

*Breidden Hill Inlier, Welsh Borderlands—Site mean statistics af cleaned and tilt corrected*

| Site No. | Cleaning fields (Oe) | $N$ (samples) | $R$ | $k$ | $\alpha_{95}$ | $D$ | $I$ | Palaeomagnetic pole (°N) | Palaeomagnetic pole (°E) |
|---|---|---|---|---|---|---|---|---|---|
| (a) | Breidden Hill dolerite intrusion | | | | | | | | |
| 1 | 50–100 | 3 | 2·9 | 113 | 12 | 338 | −18 | 25 | 202 |
| 2 | 50–100 | 3 | 2·8 | 12 | 36 | 350 | −43 | 12 | 187 |
| 3 | 200 | 11 | 10·9 | 73 | 5 | 340 | −63 | −9 | 192 |
| 4 | 200–600 | 10 | 9·8 | 46 | 7 | 29 | −75 | −26 | 162 |
| 5 | 100 | 9 | 8·5 | 16 | 13 | 335 | −65 | −12 | 194 |
| 6 | 100–200 | 9 | 8·9 | 108 | 5 | 334 | −46 | 7 | 200 |
| 7 | 50 | 7 | 6·9 | 44 | 9 | 333 | −44 | 8 | 201 |
| | Group mean of seven sites: | | | | | | | | |
| | | 7 | 6·6 | 16 | 18 | 341 | −51 | 4 | 193 |
| | | | | | | | ($d\Psi = 14{\cdot}6$, $d\chi = 21{\cdot}5°$ | | |
| (b) | Dolerite sill, Barreg Bank | | | | | | | | |
| 13 | 100–200 | 9 | 8·6 | 21 | 11 | 309 | −51 | −5 | 219 |
| (c) | Caradocian tuffs, Upper Volcanic Group | | | | | | | | |
| 8 | 100 | 5 | 3·6 | 3 | 55 | 192 | 35 | −17 | 345 |
| 10 | 200–100 | 5 | 4·9 | 37 | 12 | 164 | 44 | −10 | 11 |
| (d) | Moel-y-Golfa andesite | | | | | | | | |
| 9 | 200 | 13 | 11·9 | 11 | 13 | 290 | −63 | −24 | 224 |
| 11 | 300–600 | 8 | 7·2 | 9 | 20 | 342 | −78 | −31 | 185 |
| 12 | 300–500 | 7 | 4·2 | 2 | 55 | 357 | −20 | 27 | 180 |

Barreg Bank gives a direction of magnetization close to the Breidden Hill intrusion after tilt correction.

Sites 8 and 10 from fine and course-grained tuffs respectively of the Upper Volcanic Group show only moderate stability of af cleaning but have directions of magnetization approximately anti-parallel to those of the Breidden Hill intrusion. Four samples from site 10 possess a soft secondary component of magnetization, removed by af treatment above 200 oe, of $D = 42°$, $I = 35°$ ($k = 17$) and representative of Siluro-Devonian palaeomagnetic field directions for Britain (Briden, Morris & Piper 1973). Samples from three sites in the Moel-y-Golfa andesite show similar behaviour with slight to moderate falls in moment on demagnetizing to 1400 oe, and although individual samples show high stability, for reasons which are not clear within-site grouping is only moderate for sites 9 and 11 in quarry exposures and poor for site 12 from a surface exposure. The two better-grouped sites have directions of magnetization approximately antiparallel to those of underlying tuffs (sites 8 and 10) but similar to the unaltered sites in the Breidden Hill intrusion. This apparent reversal of magnetic field polarity is seen to be in the sense $R \rightarrow N$ by reference to the apparent polar wander curve for Britain for Palaeozoic time (Briden *et al.* 1973), and belongs to the Climacograptus *peltifer* zone of the Caradocian.

The Upper Ordovician pole positions derived from this study (Table 1) lie close to poles of Lower and Middle Ordovician age from elsewhere in Britain (Fig. 1(c)).

*Sub-department of Geophysics,*
*Oliver Lodge Laboratory,*
*University of Liverpool*
*P.O. Box* 147,
*Liverpool L*69 3*BX.*



Downloaded from https://academic.oup.com/gji/article/43/3/1013/819255 by Curtin University Library user on 26 May 2026

## References

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides VI—Regional and Global Implications, *Geophys. J. R. astr. Soc.*, **33**, 107–134.

Pocock, R. W. & Whitehead, T. H., 1948. *British Regional Geology: The Welsh Borderlands*, (2nd Ed.), Her Majesty's Stationery Office, London 82 pp.

Watts, W. W., 1925. The geology of South Shropshire, *Proc. geol. Assoc.*, **36**, 321–363.

Downloaded from https://academic.oup.com/gji/article/43/3/1013/819255 by Curtin University Library user on 26 May 2026